Math Inquirers

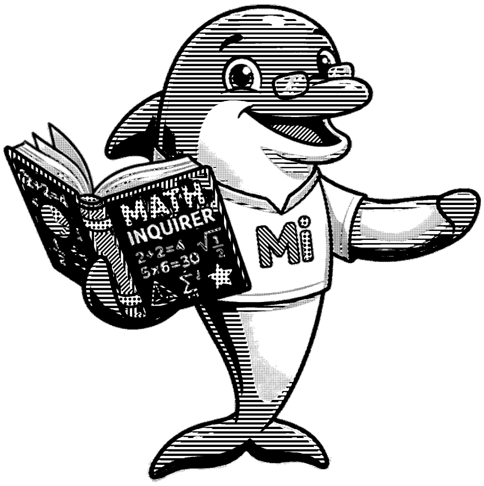
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Workbook

Book 2

Lessons 6 to 10


MATH
INQUIRER
2+2=4
5×6=30
Mi

My name:

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Math Inquirers Level 1, Term 1 - Workbook, Book 2

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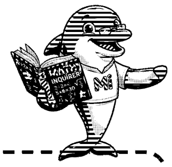
Story Steps

Kit pages (cut-outs)

Some lessons end with a kit page: tens bars, number lines, cards and so on.
Each kit page has a blank back, so you can cut or tear it out without losing any work.
Tear along the dashed line in the left margin. If the page says cut out, a grown-up cuts the pieces; keep them

TENS JUMP, ONES HOP!

Name ______________________________ **Date** ______________

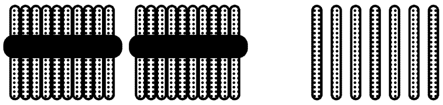


After question 3 you may choose: carry on, or try **Think harder**.

Use your kit or a drawing. A bundle of 10 is a bar in your kit. Say your answer out loud, then write it.

No.S1

These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.



(a) Now Mei has _______ bundles of ten.

(b) She also has _______ single sticks.

(c) Mei has _______ sticks altogether.

No.S2

In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?

There are _______ sticks altogether.

No.S3

Jump forward. Fill in the blanks.

(a) 23 + 10 = _______

(b) 23 + 40 = _______

(c) 51 + 30 = _______

(d) 7 + 60 = _______

ADDITION AND SUBTRACTION: TENS JUMP, ONES HOP

Start here

Name ______________________ **Date** __________

No.S4

Jump or hop. Fill in the blanks.

(a) 34 - 10 = _______

(b) 58 - 20 = _______

(c) 58 - 3 = _______

(d) 62 + 7 = _______

No.S5

Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?

There are _______ pencils left.

No.S6

Ben says 45 + 20 = 47. Fill in a to c.

(a) Is Ben correct? Circle Yes or No. **Yes** **No**

(b) The right answer is _______.

(c) Ben added 2 _______ by mistake. He should add 2 _______.

No.S7

Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?

I land on _______.

Term 1 • Lesson 6 ADDITION AND SUBTRACTION: TENS JUMP, ONES HOP **Think harder**

Name ________________________________ **Date** ____________

Use your kit or draw the jumps and hops. Then write how you know. Choose any question, in any order.

No.T1

Add 23 + 45. Fill in the blanks.

(a) Jump: 23 + _______ = _______

(b) Hop: _______ + _______ = _______

(c) So 23 + 45 = _______

No.T2

Subtract 79 - 35. Fill in the blanks.

(a) Jump back: 79 - _______ = _______

(b) Hop back: _______ - _______ = _______

(c) So 79 - 35 = _______

No.T3

Both frogs start on 12. Frog A jumps 3 tens, then hops 4. Frog B hops 4, then jumps 3 tens. Fill in a to c.

(a) Frog A lands on _______.

(b) Frog B lands on _______.

(c) Is the landing number the same for both frogs? Circle Yes or No. **Yes** **No**

....

No.T4

Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?

Ali started on _______.

....

Term 1 • Lesson 6

ADDITION AND SUBTRACTION: TENS JUMP, ONES HOP

Go further

Name ______________________________ **Date** ______________

Choose any question. You do not have to finish them all.

No.G1

Mei is on 7 and must get to 97. Each time she can jump 10 or jump 20. Many answers are right. Fill in a and b.

(a) Write the numbers Mei lands on in one way.

(b) Write the numbers for a different way.

No.G2

Now Mei wants the fewest jumps from 7 to 97. She can jump 10 or jump 20. Fill in a and b.

(a) The fewest number of jumps is ______.

(b) Why can Mei not do it in 4 jumps? I know because ______.

No.G3

Start at 25. Jump forward 3 tens, jump back 1 ten, then jump forward 2 tens. Then hop and land on 69. Fill in a to c.

(a) Write each number you land on before the hop.

(b) The hop is ______ ones forward.

(c) The three jumps are the same as one jump of ______ tens forward.

Name ______________________________ **Date** ______________

Investigation: Jump road

Try this any time. You do not have to finish.

First, try this. Start at 6. Jump in tens 5 times. Write every number you land on.

Next, try this. Start at 6 and keep jumping in tens. Will you land on 50? Write Yes or No. Then try other start numbers from 0 to 50. Which start numbers can land on 50? Write what you notice.

....

....

....

....

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Add.

46 + 30 = _______

No.E2

Subtract.

(a) 74 - 30 = _______

(b) 74 - 3 = _______

No.E3

Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.

(a) After the jump she is on _______.

(b) After the hop she is on _______.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: ....

The hardest part was ... ☐ knowing if it is a jump or a hop ☐ taking away tens ☐ jump, then hop ☐ the number line ☐ something else

What helped me: ☐ the bars and squares ☐ drawing the jumps ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name ______________________________ **Date** ______________

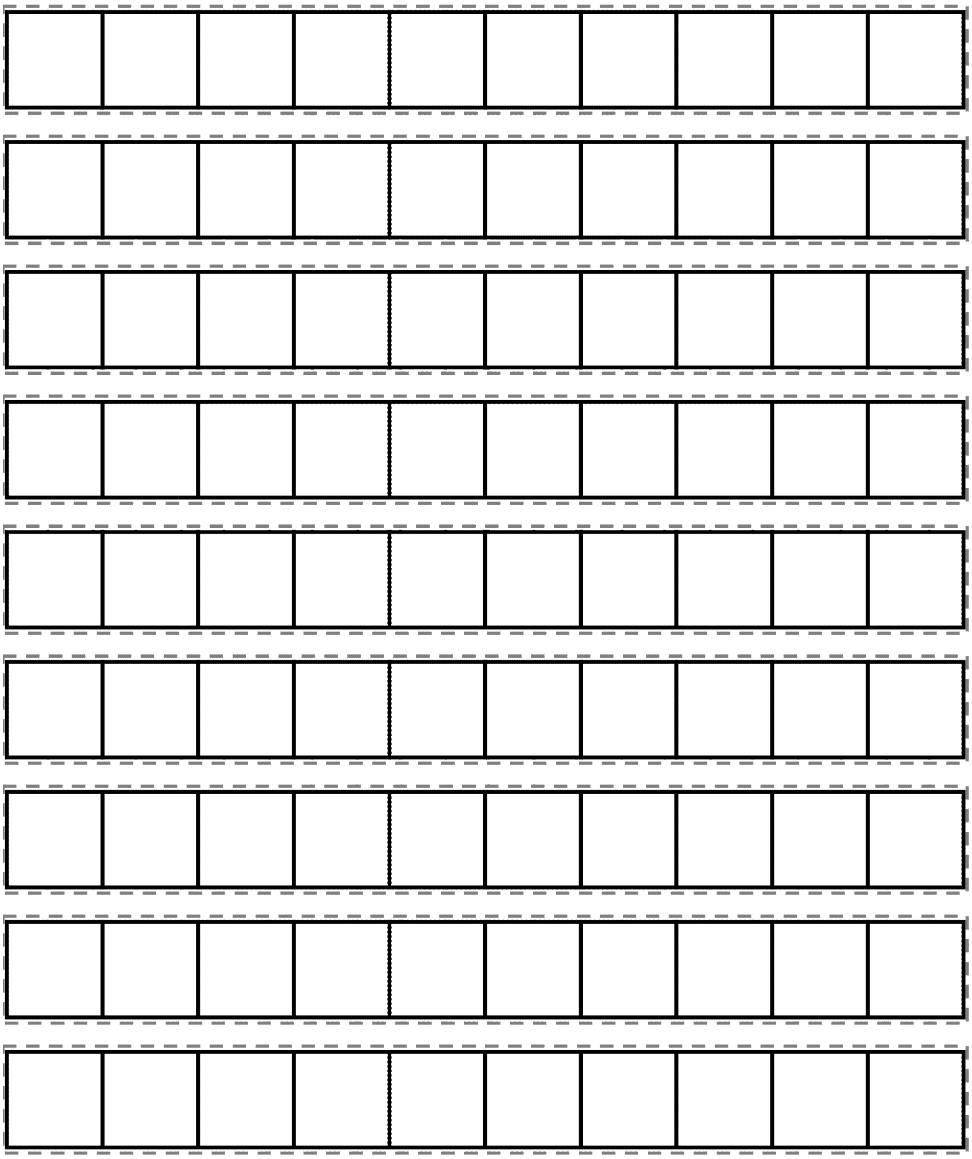
TEAR OUT here (blank on the back) - then cut out the pieces

Kit page: tens bars and ones squares

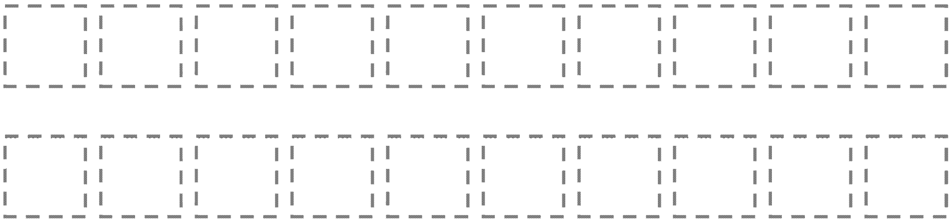
A grown-up cuts these out before the lesson. A bar is a ten. A square is a one. These are the kit from the first lesson of the term, so keep them in an envelope.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

Tens bars: cut out all 9 (each bar is a ten)



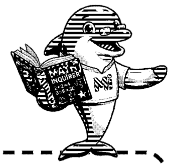
Ones squares: cut out all 20 (each is one one)



Term 1 • Lesson 7 ADDITION AND SUBTRACTION **Start here**

TAKE AWAY, COUNT UP!

Name ______________________________ **Date** ______________

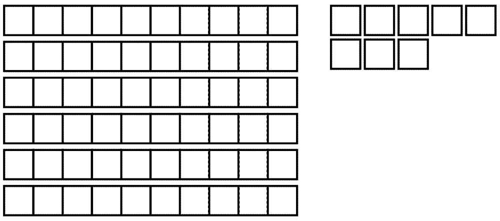


After question 3 you may choose: carry on, or try **Think harder**.

Use your kit, a drawing or your fingers. Say your answer out loud, then write it.

No.S1

Here is 68 as bars and squares. Cross out 2 bars. Then fill in the blanks.



(a) _______ tens and _______ ones are left.

(b) 68 - 20 = _______

No.S2

Subtract. Use your kit if you like.

(a) 57 - 20 = _______

(b) 57 - 4 = _______

(c) 86 - 30 = _______

(d) 86 - 5 = _______

No.S3

Find 68 - 25. Fill in the blanks.

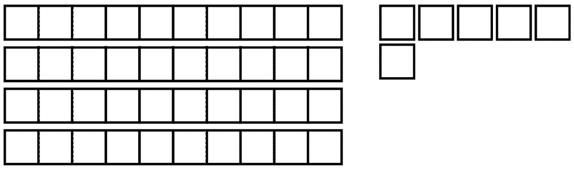
(a) 68 - 20 = _______

(b) 48 - 5 = _______

Name ______________________________ **Date** ______________

No.S4

Subtract. Then add to check.



(a) 46 - 12 = _______

(b) 34 + 12 = _______

No.S5

Mei has 45 stickers. She gives away 13. How many stickers are left? Draw a bar for each ten and a square for each one.

Mei has _______ stickers left.

No.S6

Kumar says 59 - 20 = 57.

(a) Is Kumar right? Circle Yes or No. **Yes** **No**

(b) 59 - 20 = _______

No.S7

A bus has 64 seats. 21 children are sitting. How many seats are empty?

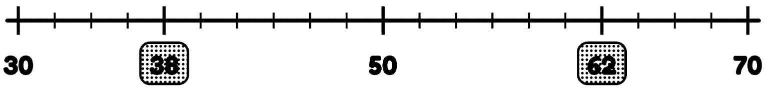
_______ seats are empty.

Name ______________________ **Date** __________

Use your kit, draw a picture or count up. Then write how you know. Choose any question, in any order.

No.T1

Mei has 62 beads. Ravi has 38 beads. Find how many more beads Mei has than Ravi.



Mei has ______ more beads.

No.T2

Find 83 - 77. Show how you found it.

83 - 77 = ______

No.T3

Ali has 24 stickers and Ben has 31 stickers. Then each child gets 10 more stickers. Find how many more stickers Ben has than Ali now.

Ben has ______ more stickers.

No.T4

A book has 71 pages. Ravi has read 46 pages. How many pages are left?

______ pages are left.

Name ________________ **Date** ________

Choose any question. You do not have to finish them all.

No.G1

Fill in the blanks. Circle the quicker way for each.

(a) 71 - 68 = _______ **take away** **count up**

(b) 71 - 20 = _______ **take away** **count up**

(c) I know because _______.

No.G2

A crate has 100 eggs. Ali sells 64 of them. How many eggs are left? Find the answer two ways.

(a) By taking away: _______ eggs are left.

(b) By counting up: _______ eggs are left.

(c) Do the two answers match? Circle Yes or No. **Yes** **No**

No.G3

Mei says 83 - 47 and 86 - 50 have the same answer.

(a) 83 - 47 = _______

(b) 86 - 50 = _______

(c) Is Mei correct? Circle Yes or No. **Yes** **No**

(d) I know because _______.

Name ______________________ **Date** __________

Investigation: When is counting up quicker?

Try this any time. You do not have to finish.

First, try this. Find 52 - 49 and 52 - 10. Try take away and count up for each one. Which way is quicker for each?

Next, try this. Write two subtractions where counting up is quicker, and two where taking away is quicker. Many answers are right. Then finish: Counting up is quicker when ___.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Subtract.

58 - 24 = _______

No.E2

Fill in the blanks.

(a) 57 + _______ = 62

(b) 62 - 57 = _______

No.E3

Find 90 - 87. Circle the way you used: take away or count up.

90 - 87 = _______

take away **count up**

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

The hardest part was ... ☐ taking away tens and ones ☐ hopping to the next ten ☐ counting up in jumps ☐ choosing a tool ☐ something else

What helped me: ☐ bars and squares ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name ____________________ **Date** ____________

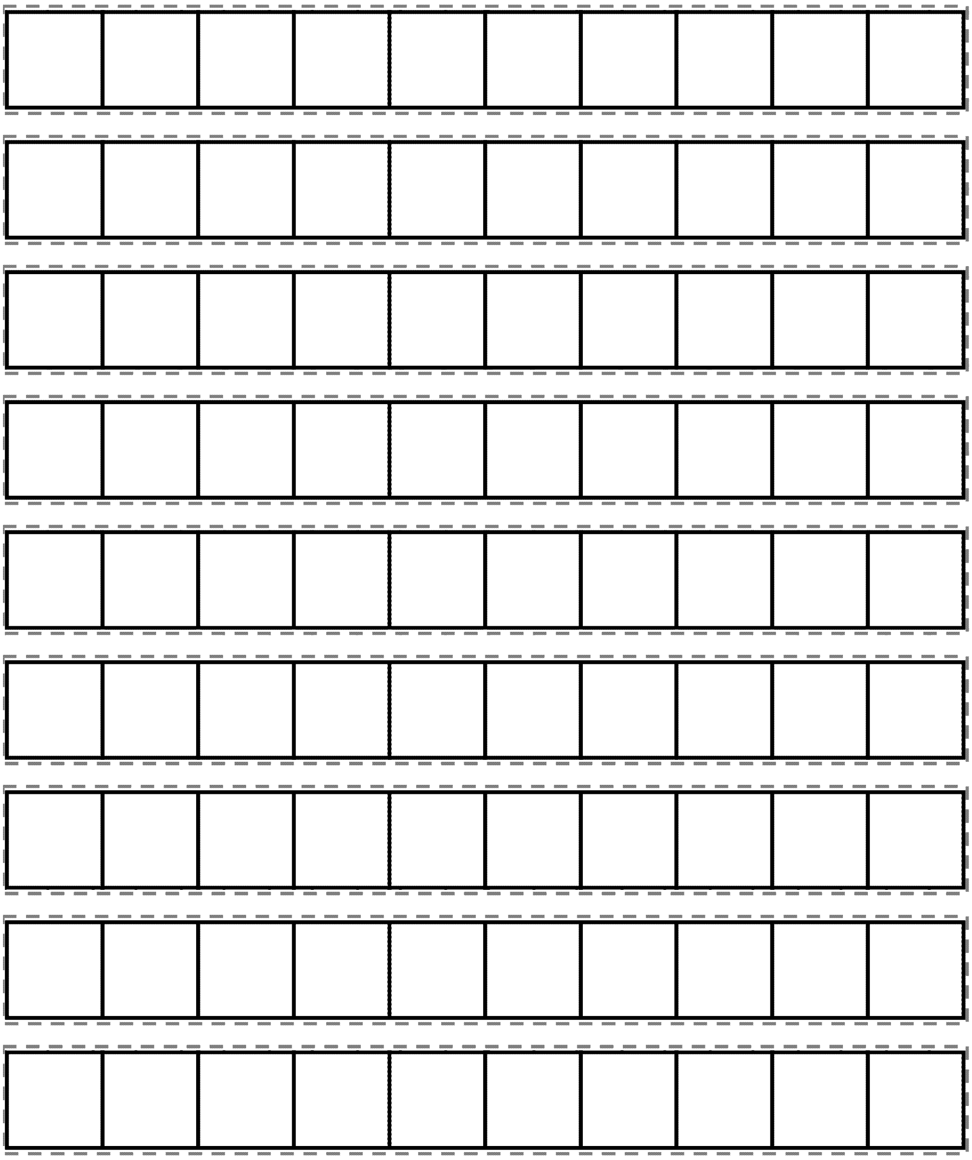
Kit page: tens bars and ones squares

A grown-up cuts these out before the lesson. A bar is a ten. A square is a one. Keep them in an envelope.

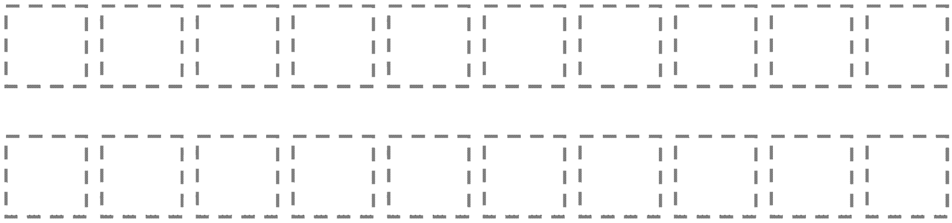
Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

TEAR OUT here (blank on the back) - then cut out the pieces

Tens bars: cut out all 9 (each bar is a ten)



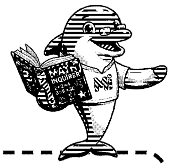
Ones squares: cut out all 20 (each is one one)



Term 1 • Lesson 8 ADDITION AND SUBTRACTION: BALANCE **Start here**

BALANCE AND SWAP!

Name ______________________ **Date** __________



After question 3 you may choose: carry on, or try **Think harder**.

Use your plates or a drawing. Say it out loud, then write it.

No.S1

Fill in the missing numbers.

(a) 5 + 3 = 4 + ☐

(b) 6 + 4 = 5 + ☐

(c) 7 + 3 = ☐ + 7

(d) 2 + 8 = 8 + ☐

No.S2

Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

She adds ☐ more buttons

No.S3

Siti writes 6 + 3 = 5 + 9.

(a) Is it balanced? Circle Yes or No. **Yes** **No**

(b) Fill in the missing number: 6 + 3 = 5 + ☐

Term 1 • Lesson 8

ADDITION AND SUBTRACTION: BALANCE

Start here

Name ______________________________ **Date** ______________

No.S4

7 + 5 = 12. Fill in the missing numbers.

(a) 6 + ☐ = 12

(b) 5 + ☐ = 12

(c) 4 + ☐ = 12

No.S5

Fill in the missing numbers.

(a) 20 + 30 = 40 + ☐

(b) 50 + 20 = 60 + ☐

(c) 40 + 20 = 30 + ☐

(d) 30 + 30 = 20 + ☐

No.S6

Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?

Ben has ☐ blue pens.

No.S7

Fill in the missing number.

35 + 12 = ☐ + 2

Name ______________________ **Date** __________

Use your plates or draw a picture. Then write how you know. Choose any question, in any order.

No.T1

Mei says 56 + 17 and 54 + 19 have the same total. Ali says 54 + 19 has the greater total. Do not add.

(a) Who is right? Circle Mei or Ali.

(b) I know because _______.

....

....

No.T2

Do not add. Circle LEFT or RIGHT for the greater total, or circle SAME.

(a) 29 + 14 and 30 + 13: LEFT / RIGHT / SAME

(b) 53 + 20 and 50 + 23: LEFT / RIGHT / SAME

(c) 26 + 15 and 28 + 16: LEFT / RIGHT / SAME

No.T3

Fill in the missing number. Then find the total.

(a) 49 + 26 = 50 + ☐

(b) The total is ☐.

....

No.T4

Ben writes 24 + 17 = 26 + 19.

(a) Is it balanced? Circle Yes or No. **Yes** **No**

(b) Change the 19 to ☐.

....

Term 1 • Lesson 8

ADDITION AND SUBTRACTION: BALANCE

Go further

Name ______________________ **Date** __________

Choose any question. You do not have to finish them all.

No.G1

Fill in the missing numbers so that all three sums have the same total.

19 + 28 = 20 + ☐ = ☐ + 30

No.G2

Fill in the blanks so both sides are the same. Write 3 different pairs. Many answers are right.

(a) ☐ + 5 = 9 + ☐

(b) ☐ + 5 = 9 + ☐

(c) ☐ + 5 = 9 + ☐

(d) I notice _______.

No.G3

Mei has 28 stickers on a page and 17 in her pocket. Ali, Ben and Kumar have the same total as Mei. Fill in the blanks.

(a) Ali has 30 stickers on his page. His pocket has ☐ stickers.

(b) Ben has 25 stickers on his page. His pocket has ☐ stickers.

(c) Kumar has 12 stickers in his pocket. His page has ☐ stickers.

Name ______________________ **Date** ______________

Investigation: Keep it the same

Try this any time. You do not have to finish.

First, try this. Add 6 + 4, then 7 + 3, then 7 + 5. Which total is not 10? Then look at 6 + 4 and 7 + 3: the 6 goes up 1. What does the 4 do?

Next, try this. Pick two numbers. Add 3 to one and take 3 from the other. Then add 3 to both. Then take 2 from both. Which keep the total the same? By how much do the others change it?

....

....

....

....

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Fill in the missing number.

9 + 5 = 10 + ☐

No.E2

Is 28 + 6 = 30 + 4 balanced?

Circle Yes or No. **Yes** **No**

No.E3

Fill in the missing number.

30 + 12 = 40 + ☐

....

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: ....

The hardest part was ... ☐ knowing what = means ☐ finding the blank ☐ how the numbers moved ☐ the big numbers (Think harder) ☐ something else

What helped me: ☐ two plates and small things ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name ____________________ **Date** __________

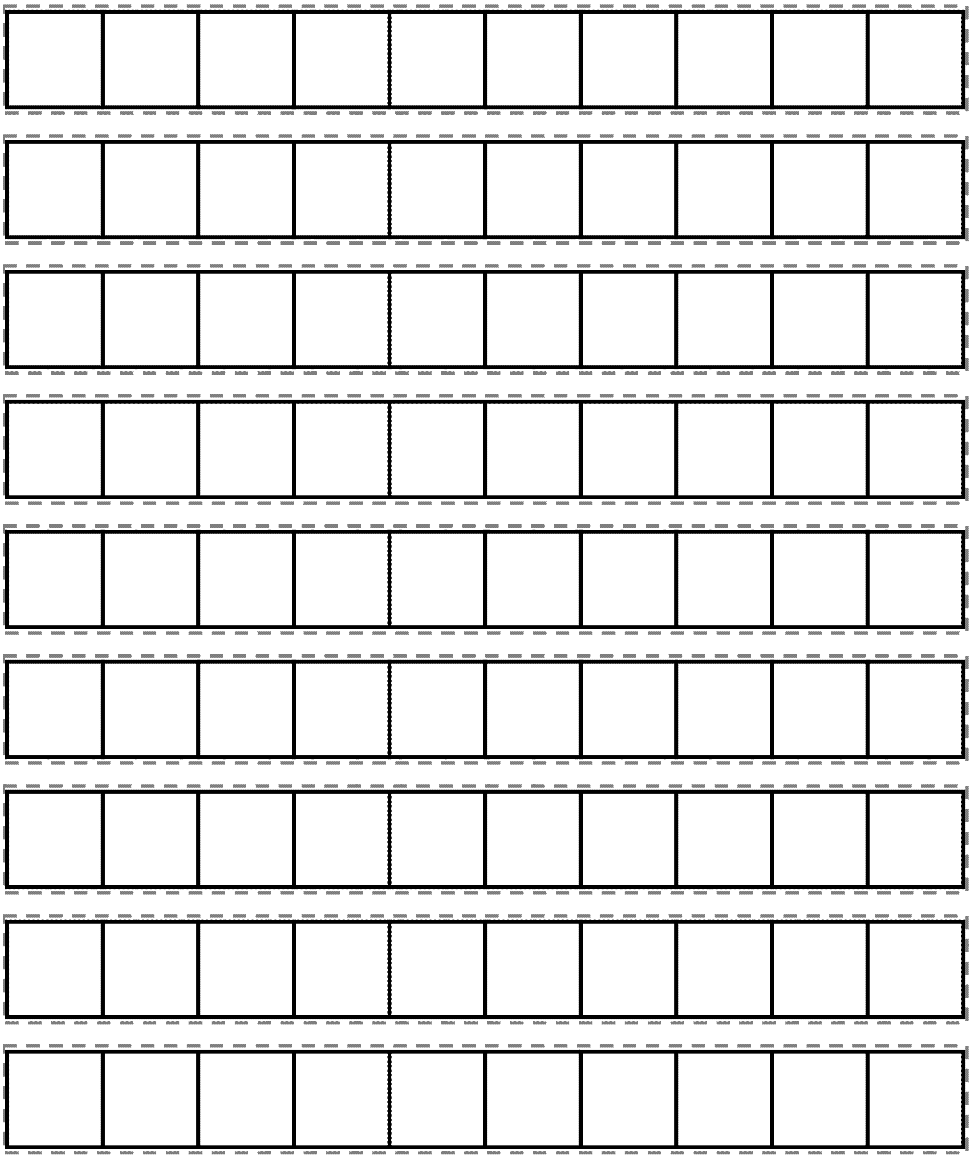
Kit page: tens bars and ones squares

A grown-up cuts these out before the lesson. A bar is a ten. A square is a one. They help with questions S5 and S7 (the tens). For the two plates, use real plates. Keep the pieces in an envelope.

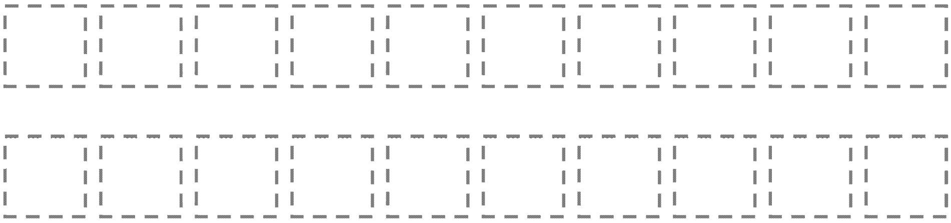
Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

TEAR OUT here (blank on the back) - then cut out the pieces

Tens bars: cut out all 9 (each bar is a ten)

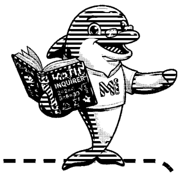


Ones squares: cut out all 20 (each is one one)



DRAW AND MEASURE!

Name ______________________________ **Date** ______________



After question 3 you may choose: carry on, or try **Think harder**.

Use your ruler and a pencil. Say your answer out loud, then write it.

No.S1

In the box, draw a line 7 cm long. Start at the 0 mark.

Your line ends at the ☐ mark.

No.S2

In the box, draw a line 9 cm long. Ali's line is 4 cm long.

Your line is ______ cm longer than Ali's line.

No.S3

Measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.

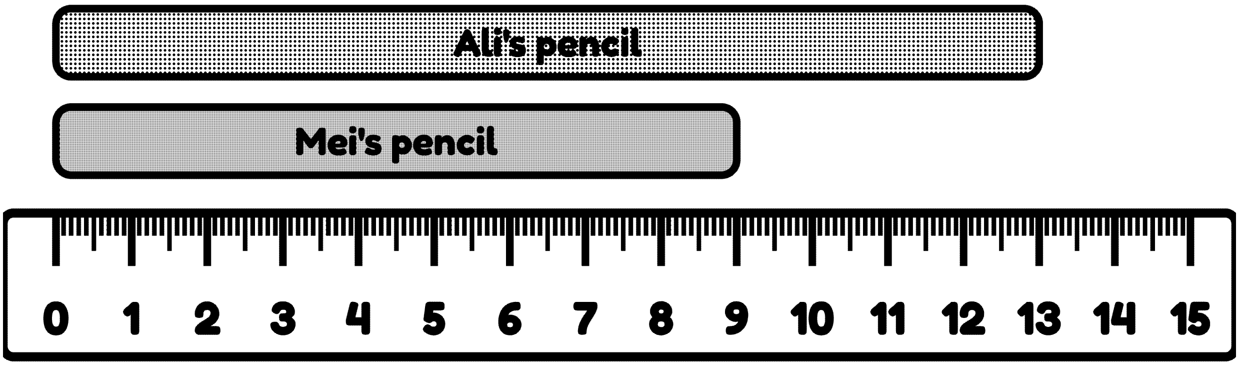
(a) My guess: ______ cm

(b) I measured: ______ cm

Name ____________________ **Date** __________

No.S4

Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.


Ali's pencil
Mei's pencil
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

(a) Ali's pencil is ______ cm long.

(b) Mei's pencil is ______ cm long.

(c) Ali's pencil is ______ cm longer than Mei's.

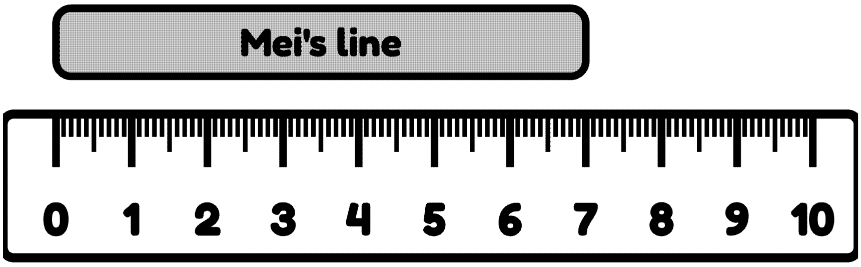
No.S5

Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.

Your line is ______ cm long.

No.S6

Look at Mei's line on the ruler. Mei says it is 8 cm long.


Mei's line
0 1 2 3 4 5 6 7 8 9 10

(a) Is Mei correct? Circle Yes or No. **Yes** **No**

(b) Her line is ______ cm long.

Name ______________________ **Date** ____________

Use your ruler and drawings to help. Choose any question, in any order.

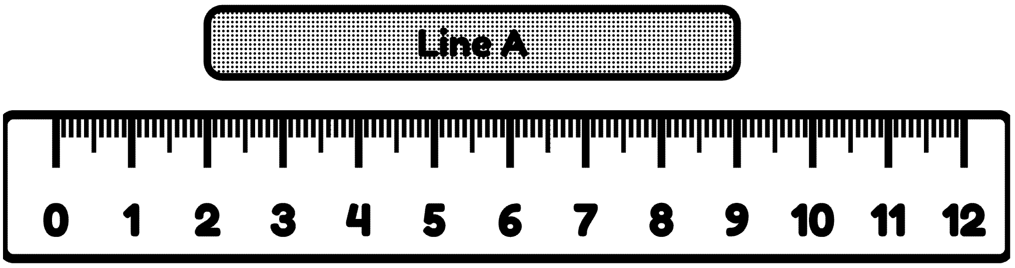
No.T1

Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.

The whole path is ______ cm long.

No.T2

Look at line A on the ruler. Line B starts at the 3 mark and is 6 cm long.

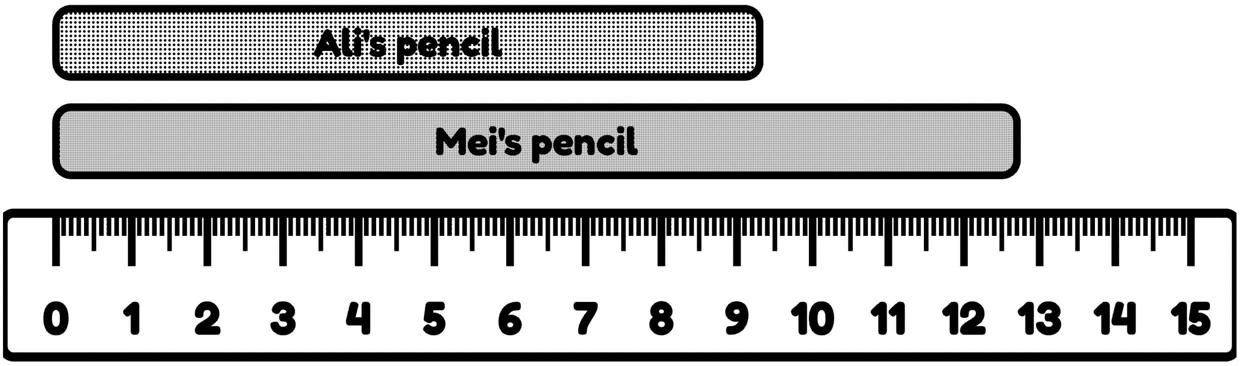

Line A
0
1
2
3
4
5
6
7
8
9
10
11
12

(a) Line A is ______ cm long.

(b) Line B ends at the ______ mark.

No.T3

Look at the ruler. Both pencils start at the 0 mark. Each tip is between two marks.


Ali's pencil
Mei's pencil
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

(a) To the nearest cm, Ali's pencil is ______ cm long.

(b) To the nearest cm, Mei's pencil is ______ cm long.

(c) Mei's pencil is ______ cm longer than Ali's.

Name ______________________________ **Date** ______________

Choose any question. You do not have to finish them all.

No.G1

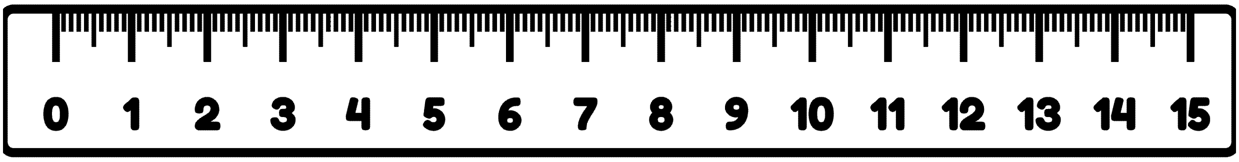
A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?

Piece C is _______ cm long.

No.G2

A snail crawls along a long ruler from the 4 mark to the 13 mark. Then it crawls back 5 cm.

The snail's way forward



(a) The snail is now at the _______ mark.

(b) The snail has crawled _______ cm in all, counting the way back.

No.G3

Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.

My 3 lengths: _______ cm, _______ cm and _______ cm.

Name ______________________________ **Date** ______________

Investigation: Does the start mark matter?

Try this any time. You do not have to finish.

First, try this. Draw a line 6 cm long from the 2 mark. Where does it end?

Next, try this. Try 3 more start marks, 6 cm each time. Lines 1 to 3: write the start and end marks. Line 4: I notice ...

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

In the box, draw a line 6 cm long. Start at the 0 mark.

Your line ends at the ☐ mark.

No.E2

Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

The pencil is ______ cm longer than the rubber.

No.E3

A line starts at the 3 mark and ends at the 10 mark. How long is the line?

The line is ______ cm long.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: ....

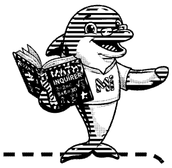
The hardest part was ... ☐ starting at the 0 mark ☐ drawing a line exactly ☐ a line that starts at another mark ☐ adding the pieces of a path ☐ something else

What helped me: ☐ my ruler ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

STORY STEPS: READ IT TWICE!



Name ______________________ **Date** __________

After question 3 you may choose: carry on, or try **Think harder**.

Use counters or a drawing. Read the story twice. Say your answer out loud, then write it.

No.S1

Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards.

(a) Step 1: 30 - 20 = ☐

(b) Step 2: ☐ + 6 = ☐

(c) Ben has ☐ cards now.

No.S2

Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step.

(a) Step 1: _______

(b) Step 2: _______

(c) Mei has _______ stickers now.

No.S3

Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the 2 number sentences that fit the story.

(a) 25 + 10 = 35

(b) 25 - 10 = 15

(c) 35 - 3 = 32

(d) 35 + 3 = 38

Ravi has _______ pencils now.

Term 1 • Lesson 10 STORY STEPS **Start here**

Name ______________________________ **Date** ______________

No.S4

Write + or - in each box.

(a) Ali has 8 shells. He finds 5 more shells. 8 ☐ 5 = 13

(b) Siti has 9 sweets. She eats 3 sweets. 9 ☐ 3 = 6

(c) Mei has 10 stickers. She gives away 4 stickers. 10 ☐ 4 = 6

(d) 12 children stand in a line. 6 more join. 12 ☐ 6 = 18

No.S5

Ali has 23 marbles. Ben has 10 more than Ali. How many marbles do they have altogether? Use counters if you like.

They have ______ marbles altogether.

No.S6

A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 + 10 = 40 for Step 1.

(a) Is Siti right? Circle Yes or No. **Yes** **No**

(b) The bus now has ______ children.

Name ______________________________ **Date** ______________

Use counters or draw a picture to help. Choose any question, in any order. Pages 3 to 5 can be finished at home.

No.T1

Ravi has 15 marbles. He has 3 more marbles than Mei.

(a) Mei has _______ marbles.

(b) They have _______ marbles altogether.

No.T2

Mei is in a line to buy lunch. There are 6 children in front of her and 4 children behind her. How many children are in the line?

There are _______ children in the line.

No.T3

Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles does Siti have?

Siti has _______ marbles.

No.T4

Ali has 12 stickers. He gives 7 stickers to Mei. Then he gives 8 stickers to Ben.

(a) After giving 7 to Mei, Ali has _______ stickers left.

(b) Can Ali give 8 stickers to Ben? Circle Yes or No. **Yes** **No**

Name ______________________________ **Date** ______________

Choose any question. Write one number sentence for each step. You do not have to finish them all.

No.G1

A shop has 60 pencils. It sells 24 pencils on Monday and 15 pencils on Tuesday. On Wednesday, a box of 30 pencils comes in. Write one number sentence for each step.

(a) Step 1: _______

(b) Step 2: _______

(c) Step 3: _______

(d) The shop has _______ pencils now.

No.G2

Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End your story with a question. Many stories are right.

No.G3

Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. Write one number sentence for each step. How many shells does Ben have?

(a) Step 1: _______

(b) Step 2: _______

(c) Step 3: _______

(d) Ben has _______ shells.

Name ______________________________ **Date** ______________

Investigation: Does the order matter?

Try this any time. You do not have to finish.

First, try this. Start at 20. Add 10, then take away 5. Start again at 20. Take away 5 first, then add 10. Is the answer the same?

Next, try this. Pick your own numbers. Try both orders. Is the answer always the same? When can one order not work?

....

....

....

....

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

A bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now?

The bus now has _______ children.

No.E2

Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have?

Mei has _______ stickers.

No.E3

Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?

Ali has _______ beads.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: ....

The hardest part was ... ☐ reading the story twice ☐ add or take away ☐ more than, fewer than ☐ starting Step 2 from Step 1 ☐ something else

What helped me: ☐ counters ☐ drawing it ☐ reading twice ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ counters ☐ checking ☐ asking ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name ______________________ **Date** __________

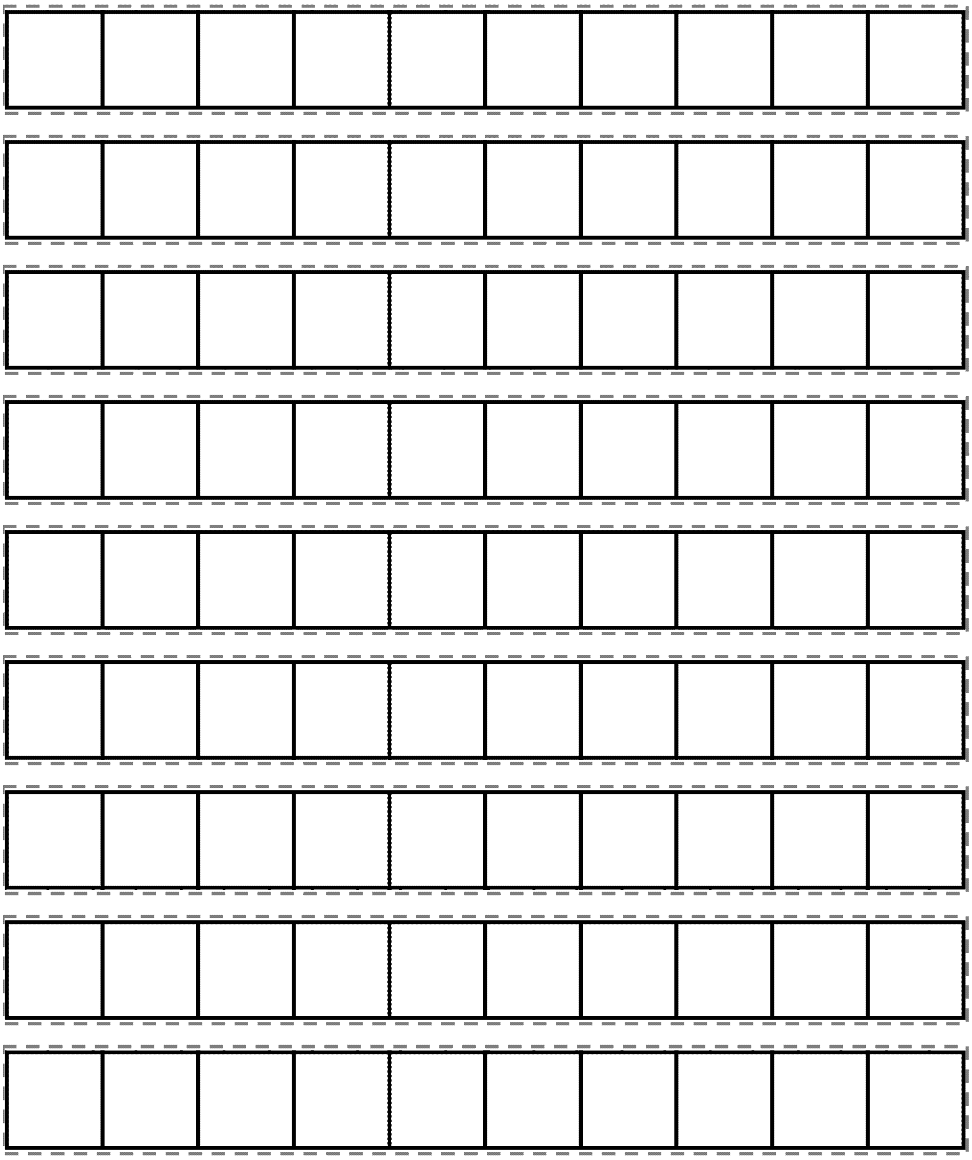
Kit page: tens bars and ones squares

Cut these out first. The squares make good counters. A bar is a ten.

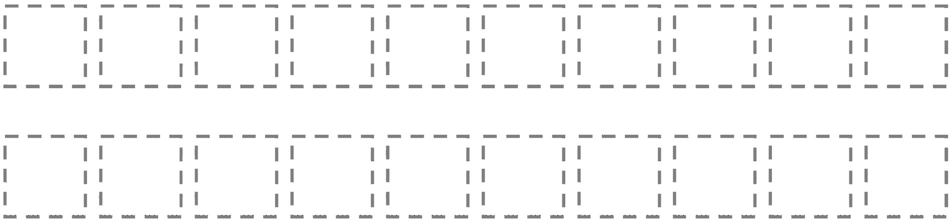
Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

TEAR OUT here (blank on the back) - then cut out the pieces

Tens bars: cut out all 9 (each bar is a ten)



Ones squares: cut out all 20 (each is one one)

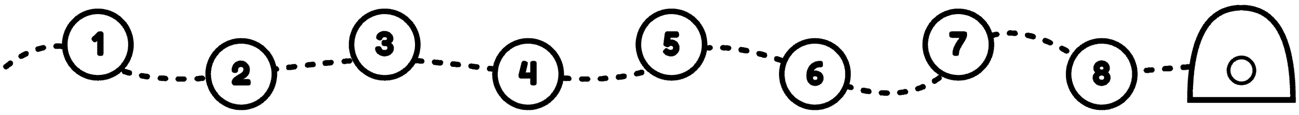


MI'S CHALLENGE QUEST

Name ______________________________ **Date** ______________

Mi says

- **Mi is swimming to the Pearl Cave and needs your help at 8 gates.**
- **Work on your own paper. Your quest partner sits beside you: talk about how, then write your own answer.**
- **Not a race. Skip a gate if you like. Only your grown-up and tutor see your answers.**
- **Gate 7 needs your ruler.**
- **The secret cave at the end is optional, just for fun.**

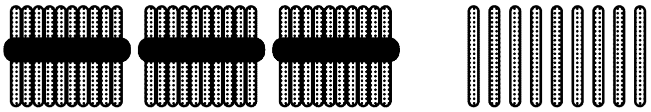


Colour a stone when you have had a go at that gate. If you skip a gate, Mi swims round it and you can still colour it.

Gates 1 and 2: Mi needs your help. Write your answers in the spaces.

Gate1

Count the sticks. Fill in the blanks.

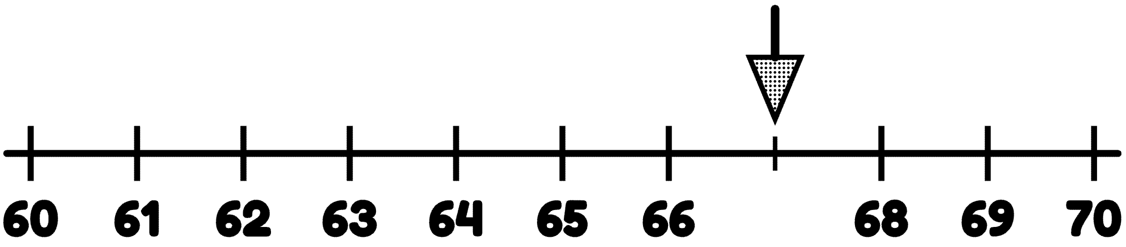


a _______ tens and _______ ones

b There are _______ sticks altogether.

Gate2

Fill in the blanks.



a Look at the number line. The arrow is pointing to _______.

b Counting in 10s: 14, 24, 34, _______

Name ______________________________

Gates 3, 4 and 5: keep swimming. Write your answers in the spaces.

Gate3

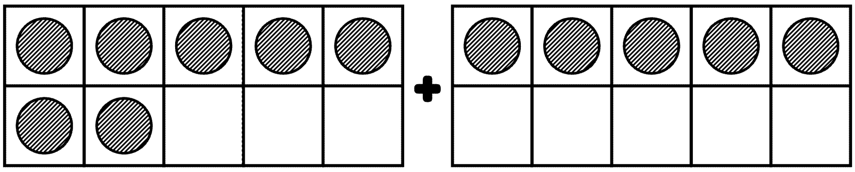
Look at the two cards. Use both cards to make the greatest number you can.

3 **8**

The greatest number is _______.

Gate4

Look at the ten frames. Fill in the missing numbers.



a 7 + 5 = ☐

b 37 + 5 = ☐

Gate5

Add or subtract.

a 52 + 30 = _______

b 74 - 31 = _______

Name ____

Gates 6 and 7: nearly there. Use your ruler for Gate 7.

Gate6

Fill in the missing number.

a 8 + 4 = 6 + ☐

b 60 + 10 = 50 + ☐

Gate7

Use your ruler.

a Draw a line 6 cm long. Start at the dot.

●

b A line 4 cm longer than your line is ______ cm long.

Name ____________________

Gate 8 and the secret cave: show your thinking. Write your answers in the spaces.

Gate8

Mei has 35 stickers. Ali gives her 10 more stickers. Then she gives 3 stickers to Ben. Write one number sentence for each step. The last number you write is how many stickers Mei has now.

a Step 1: ____________________

b Step 2: ____________________

Secret cave (optional)

Ali is in a line. There are 5 children in front of him and 3 children behind him. How many children are in the line?

Draw if you like.

There are ______ children in the line.

You may skip this one. Mi will swim round it.

Well done for trying your best!

Print-size bar (for the grown-up): this bar must measure exactly 10 cm. If it does not, print again at 100% (actual size).